

Project Proposal

Title: Predicting Employee Attrition for Proactive HR Retention

1. Problem. Employee turnover is costly in recruitment, training and lost productivity. This project predicts whether an employee will leave, so HR can step in early. Examples are reducing overtime, reviewing pay or planning promotions.

2. Dataset. The IBM HR Analytics Employee Attrition & Performance dataset (Kaggle, pavansubhasht) has 1,470 employees, 34 features and one target. It covers demographics, job role, income, satisfaction ratings, tenure, travel and overtime. It is a good fit for three reasons:

- It mixes numeric, ordinal and nominal features.
- It has a real class imbalance, with 16.1% of employees leaving.
- It has skewed variables with outliers, plus constant and ID columns to remove.

The dataset is fictional and was created by IBM data scientists, so results may not carry over to real HR data.

3. Target variable. `Attrition` (Yes = 1, No = 0).

4. Task type and success metric. This is supervised binary classification. The primary metric is PR-AUC, because accuracy would be misleading: a model that always predicts "stays" scores 83.9%. The secondary metrics are recall and F1 on the "Left" class.

5. Approach.

- Hold out a stratified 20% test set first.
- Build a `Pipeline + ColumnTransformer`. Imputation, IQR winsorising, ordinal and one-hot encoding, scaling, 7 constructed features and mutual-information selection are all fitted on training folds only.
- Handle imbalance with class weights and stratified 5-fold CV.
- Use Logistic Regression as the baseline (PR-AUC 0.52 on test), then compare tree ensembles in Part 2.

6. Risks. The risks are data leakage (shown and controlled in the notebook), class imbalance, correlated features, and the synthetic nature of the data.